

PAKS, Hungary

WMO: 129500

Lat: 46.583N

Lon: 18.850E

Elev: 98

StdP: 100.15

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-9.0	-15.0	1.0	-10.7	-12.2	1.3	-7.2	6.5	4.3	5.8	2.7	1.0	350	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	34.0	21.4	32.3	20.8	30.6	20.1	22.5	31.1	21.6	30.0	20.9	28.8	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.7	25.3	19.0	13.9	24.3	18.2	13.3	23.6	66.4	31.0	63.4	30.1	60.6	28.9	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.6	4.9	4.3	-16.9	36.3	3.7	1.5	-19.6	37.4	-21.8	38.3	-23.8	39.2	-26.5	40.3		
(5)			-17.4	24.0	3.7	0.9	-20.1	24.7	-22.2	25.2	-24.3	25.7	-26.9	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	0.0	1.8	6.1	11.6	16.3	19.8	21.7	21.4	16.3	11.3	6.0	0.9
	DBStd	8.70	4.53	5.03	4.35	3.88	3.60	3.45	3.10	3.14	3.49	4.25	4.26	4.64
	HDD10.0	1155	309	232	134	28	2	0	0	0	1	37	131	282
	HDD18.3	2945	567	464	380	203	83	23	7	9	80	219	369	540
	CDD10.0	1576	0	2	13	75	196	296	363	353	189	77	11	1
	CDD18.3	324	0	0	0	1	19	68	112	103	19	1	0	0
	CDH23.3	4202	0	0	0	52	338	879	1371	1263	271	28	0	0
	CDH26.7	1675	0	0	0	8	88	336	608	559	75	1	0	0
Wind	WSAvg	1.8	1.8	2.0	2.2	2.1	1.9	1.7	1.8	1.5	1.5	1.5	1.7	1.8
Precipitation	PrecAvg	596	33	35	30	39	58	67	66	57	61	53	51	45
	PrecMax	961	81	87	124	97	194	125	154	170	167	122	135	108
	PrecMin	397	4	1	4	0	13	12	26	6	11	2	0	0
	PrecStd	129	21	26	24	23	35	36	37	35	41	34	28	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	16.7	21.7	27.9	31.3	34.5	36.6	36.6	31.6	26.1	19.4	13.3
		MCWB	9.2	10.0	12.1	16.1	18.8	21.5	21.6	21.3	19.6	17.6	13.9	10.2
	2%	DB	10.5	14.0	19.0	24.6	29.1	32.5	33.9	34.1	28.9	23.8	16.6	10.4
		MCWB	7.5	8.8	11.1	14.4	18.3	21.2	21.6	21.3	18.7	16.4	12.6	8.1
	5%	DB	8.3	11.5	16.5	22.5	27.2	30.5	32.1	32.0	26.5	21.6	14.5	8.4
		MCWB	6.0	7.6	9.9	13.4	17.5	20.5	20.9	20.6	17.7	15.3	11.4	6.6
	10%	DB	6.4	9.1	14.1	20.3	25.0	28.5	30.3	29.9	24.2	19.1	12.7	7.0
		MCWB	4.7	6.3	8.8	12.4	16.6	19.4	20.3	19.7	16.9	14.2	10.1	5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.6	13.2	17.0	20.2	23.2	23.6	23.4	20.6	18.4	14.5	10.5
		MCDB	12.5	15.4	19.4	25.9	28.8	31.6	32.1	32.0	28.6	24.7	18.3	12.9
	2%	WB	7.8	9.3	11.7	15.3	19.0	22.1	22.4	22.2	19.4	17.0	13.0	8.4
		MCDB	10.1	13.4	17.7	22.7	27.1	30.2	31.3	30.8	26.7	22.3	16.0	10.0
	5%	WB	6.3	7.9	10.6	14.2	18.0	21.1	21.6	21.4	18.5	15.8	11.6	6.9
		MCDB	8.0	10.7	15.7	20.7	25.6	28.8	30.0	29.8	24.9	20.6	14.3	8.2
	10%	WB	4.8	6.6	9.2	13.1	17.1	20.1	20.8	20.6	17.5	14.6	10.3	5.7
		MCDB	6.2	9.0	13.5	19.1	23.9	27.0	28.6	28.4	22.9	18.6	12.4	6.8
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.9	11.2	12.9	13.2	13.2	13.6	14.0	12.5	12.1	8.3	6.2
		MCDBR	10.0	13.0	15.9	17.5	17.4	16.7	17.2	17.9	16.7	15.6	11.7	8.6
		MCWBR	7.3	8.6	9.2	8.8	7.7	6.7	6.3	6.6	7.6	8.7	7.9	6.5
	5% WB	MCDBR	9.7	11.4	14.0	15.3	15.6	15.4	15.4	15.8	14.6	14.2	10.9	8.1
		MCWBR	7.3	8.0	8.3	7.9	7.2	6.4	6.1	6.3	6.9	8.2	7.6	6.4
Clear-Sky Solar Irradiance	taub	0.336	0.359	0.395	0.447	0.434	0.433	0.444	0.446	0.425	0.390	0.361	0.335	
	taud	2.365	2.297	2.229	2.094	2.196	2.232	2.202	2.206	2.229	2.315	2.348	2.369	
	Ebn at Noon	743	798	820	807	835	836	820	801	780	751	701	696	
	Edh at Noon	75	97	120	150	140	136	138	133	119	94	76	67	
All-Sky Solar Radiation	RadAvg	1.12	1.95	3.15	4.61	5.55	6.03	5.99	5.38	3.77	2.50	1.39	0.86	
	RadStd	0.14	0.29	0.43	0.43	0.53	0.45	0.45	0.53	0.43	0.26	0.13	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	N/A	N/A	+0.86	+0.66	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+0.62	N/A	N/A	N/A	+0.50	+0.54	-117	-172	+98	+55

Nomenclature: See separate page